CHIP: A multi-sensor dataset for 6D pose estimation of chairs in industrial settings

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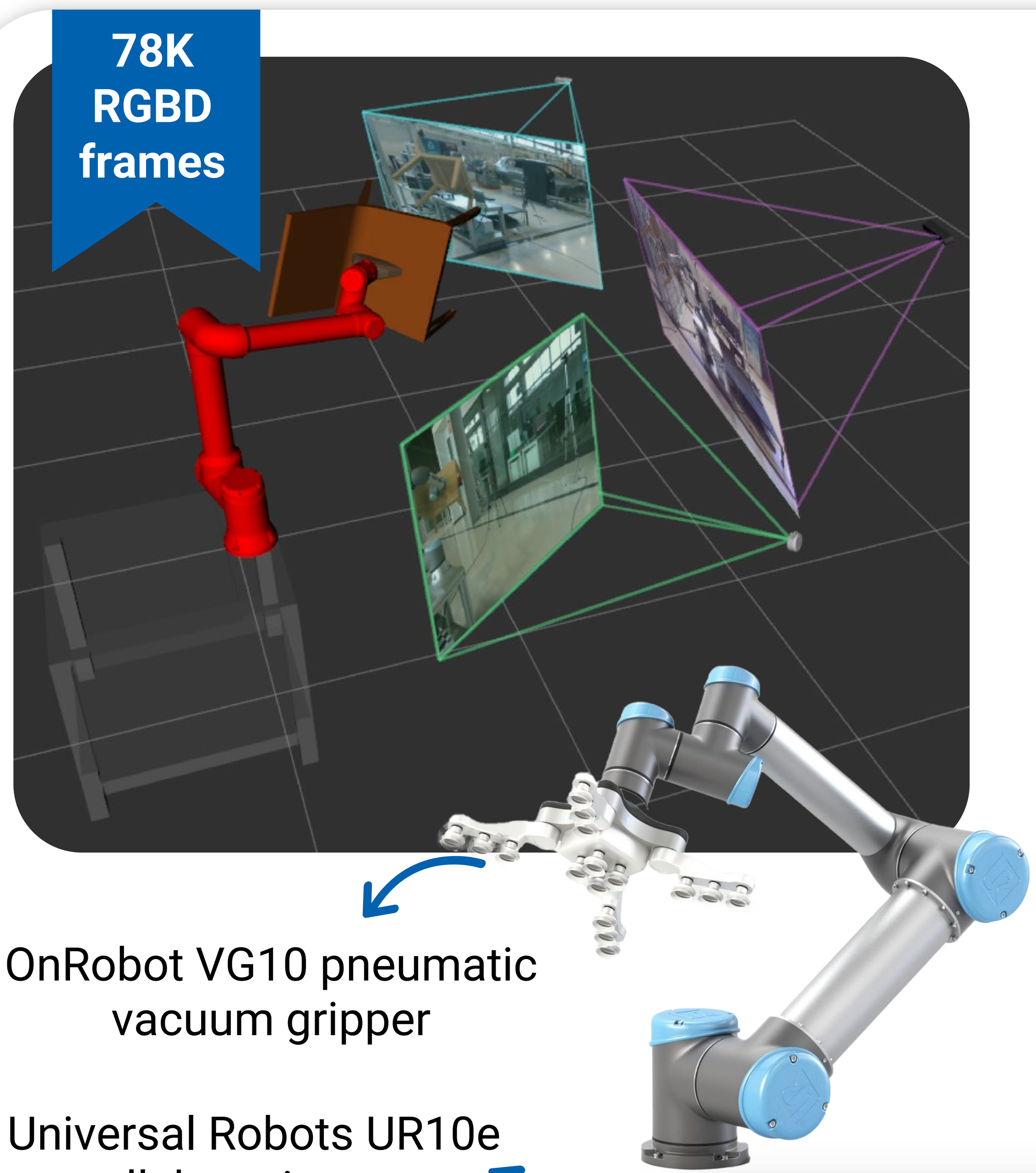
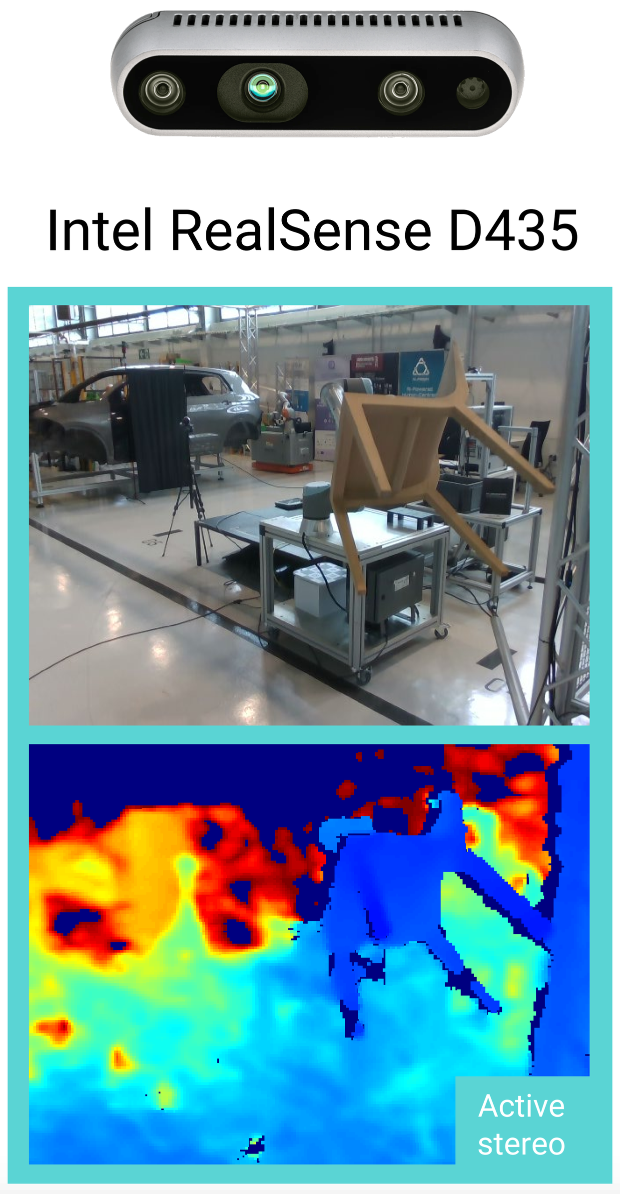


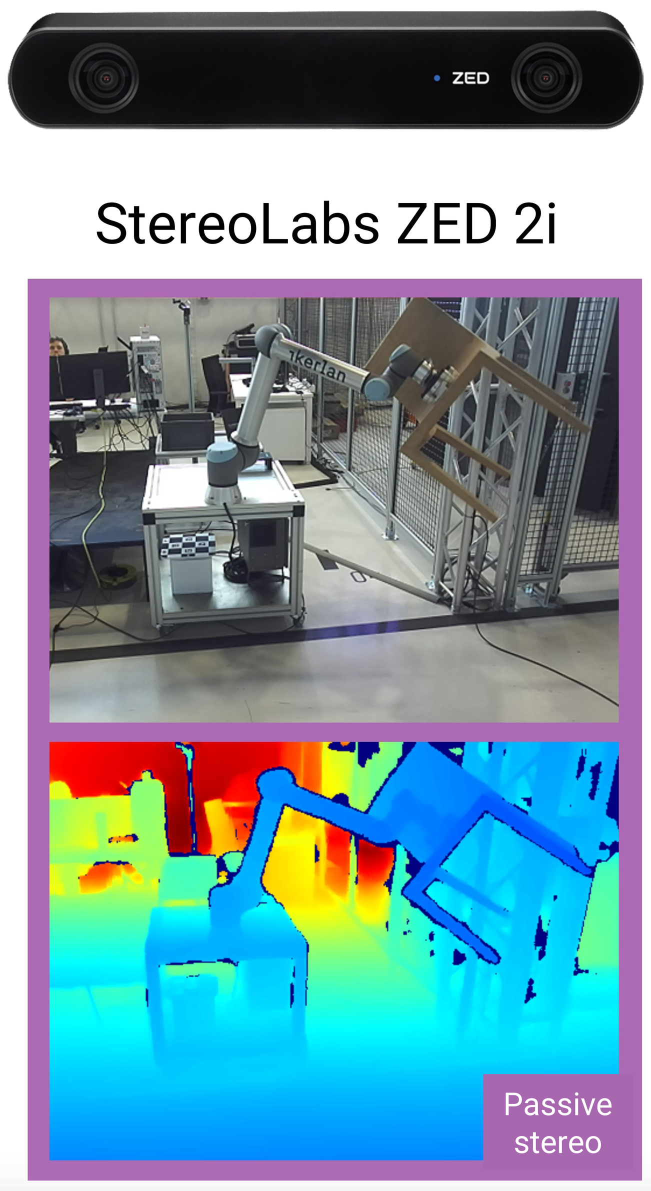
OnRobot VG10 pneumatic vacuum gripper

Universal Robots UR10e collaborative arm

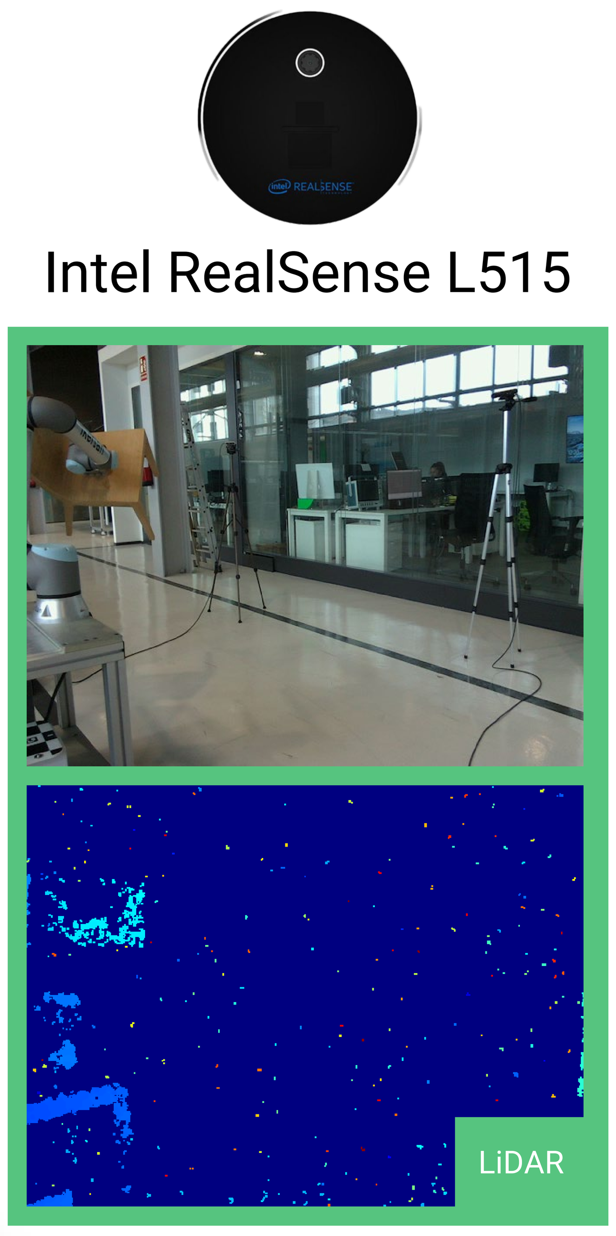
Intel RealSense D435



StereoLabs ZED 2i



Intel RealSense L515



The first 6D pose estimation dataset of chairs in challenging industrial scenarios

Why CHIP?

- Contains **large manufactured objects**, instead of small everyday items.
- Captured in a **real industrial environment** using a robotic arm.
- **Multi-sensor RGBD dataset** with different depth sensing technologies.

CHIP challenges

- **Slim and hollow chair parts** make segmentation and 6D pose estimation hard.
- Occlusions from the robotic arm, human operators, and distractor chairs introduce **realistic visual clutter**.
- Depth noise and sensor differences makes **cross-sensor generalization** challenging.

SOTA 6D pose estimators still struggle with CHIP

Method: FreeZe[1]–GeDi[2]. Metrics: rotation error (E^R) and translation error (E^T)[3].

[1] Caraffa, et al. 'FreeZe: Training-freezero-shot 6D pose estimation with geometric and vision foundation models.' ECCV, 2024.
[2] Poiesi, et al. 'Learning general and distinctive 3D local deep descriptorsfor point cloud registration.' TPAMI, 2023.
[3] Drost, et al. 'Introducing MVTec ITODD – A dataset for 3D object recognition in industry.' ICCV-W, 2017.

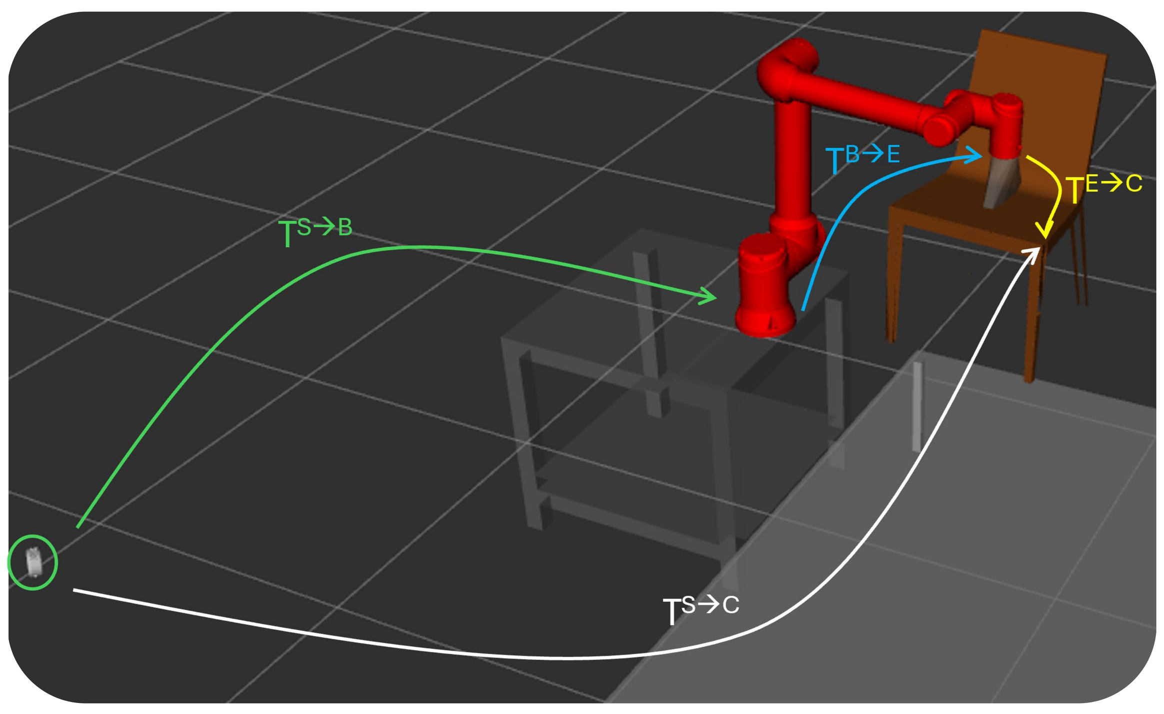
Tired of manually annotating? Let the robot handle it.

The **ground-truth 6D pose** of the chair is obtained by combining three transformations:

Sensor–base transformation from each sensor to the robot base (obtained using classical calibration)

Base–end-effector transformation from the robot base to the end-effector (obtained using robot kinematics)

End-effector–chair transformation from the end-effector to the chair (obtained using TCP)



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